Abstract

We built a predictive model for team-level goals per game (GPG) in Division I men's lacrosse and used it to generate a data-driven offensive forecast for the University of Virginia (UVA) in the 2026 season. Using a panel of team-season statistics from 2022 to 2026, we fit an XGBoost regression model on lagged offensive metrics and roster continuity features. On a 2025 backtest, the model achieves a mean absolute error of about 1.22 goals per game, which is sufficient for ranking offenses and placing teams into tiers. We then refit the model on all known seasons and forecast 2026 GPG for each team. The model projects UVA at roughly 11 goals per game, ranking 12th among offenses in our dataset. We conclude by examining the main drivers of offensive success, prior GPG, turnovers, returning production, and efficiency, and by discussing limitations and possible extensions.

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1. Introduction

The goal of our capstone project is to use data to understand and forecast offensive performance in NCAA Division I men's lacrosse, with a primary focus on the University of Virginia (UVA). In particular, we want to answer:

- How many goals per game should UVA be expected to score in the upcoming 2026 season?
- Which teams project as the best offenses nationally when we account for returning production, transfers, and prior offensive statistics?
- Which team-level features matter most for predicting future scoring?

Our original analysis plan envisioned a broad modeling framework including regularized generalized linear models for wins, win percentage, and goal differential, plus simulation-based frameworks for season outcomes. As the project progressed and we refined our scope, we chose to focus on a more targeted but high-impact question: forecasting goals per game using a flexible machine learning model.

We ultimately selected XGBoost, a gradient-boosted decision tree model, as our final predictive model for team-level GPG. The rest of this report documents our data, methods, results, and interpretations in sufficient detail for a technical audience to replicate our analysis.

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2. Data

2.1 Source and coverage

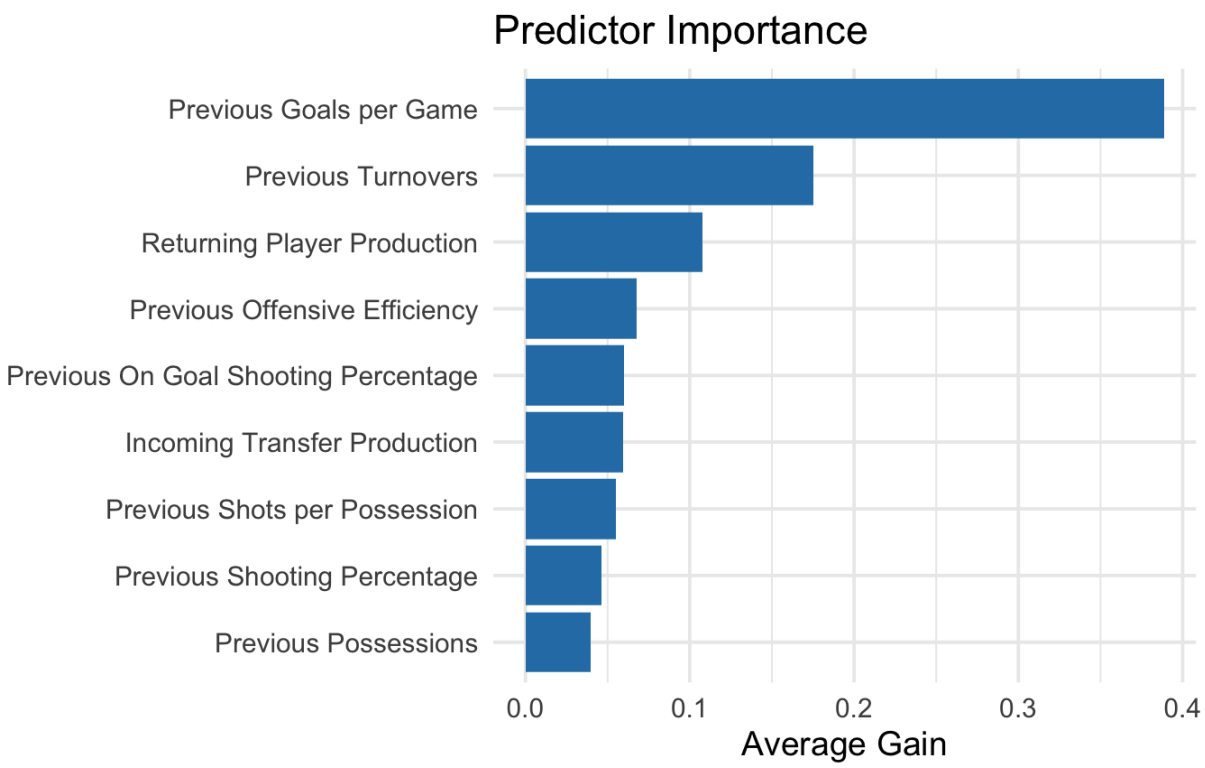
We assembled a team-season panel for Division I men's lacrosse covering seasons 2022 to 2026 for a subset of 24 programs with consistent statistics across years. In total, the final dataset (lacrosse_final.csv) contains 112 rows and 12 columns.

The fields are:

- **Team**: team name (e.g., UVA, Duke, Harvard).
- **Season**: season year (2022 to 2026).
- **Goals_g**: team goals per game in that season (continuous).
- **Lag_Goals_g**: prior-year goals per game.
- **Lag_OE**: prior-year offensive efficiency.
- **Lag_Sh**: prior-year shot volume.
- **Lag_ogSh**: prior-year shot quality (quality-adjusted shooting).
- **Lag_Poss**: prior-year possessions or pace.
- **Lag_TO**: prior-year turnover level.
- **Lag_ShPos**: prior-year shots per possession.
- **Ret_Production**: percentage of prior offensive production returning.
- **Transfer_Production**: expected offensive value from incoming transfers.

Each row represents a team-season. For 2026, the outcome Goals_g is intentionally missing, as this is what we seek to forecast.

These features are derived from a broader player-level dataset described in our analysis plan, including goals, assists, shooting percentage, turnovers, ground balls, and advanced offensive metrics, aggregated up to the team level.



2.2 Trustworthiness and cleaning

The underlying statistics are taken from LacrosseReference Pro (subscriber pages based on official game data), which we consider highly reliable. However, our data pipeline required manual extraction and transformation, which introduces risk of transcription error.

To mitigate this, we:

- Standardized team names and season formats.
- Trimmed stray spaces from column names and converted Season to integer in R.
- Performed spot checks on key fields such as goals, assists, and save percentage.
- Ensured internal consistency of derived metrics, for example quality-adjusted shooting and share-based statistics.

We judge the remaining measurement error to be small relative to the variation in team offensive performance.

2.3 Data sources

Our dataset combines information from several trusted lacrosse statistics resources. The primary source is **LacrosseReference Pro**, which provides team and player level box score

statistics and advanced offensive metrics for Division I men's lacrosse. We used these tables to obtain goals, assists, shooting statistics, turnovers, ground balls, and advanced measures such as quality adjusted shooting and expected goals added for recent seasons.

We supplemented these data with information from the **NCAA men's lacrosse statistics manual** and NCAA game box scores. The statistics manual clarified official definitions for common metrics such as goals, assists, ground balls, faceoff wins, clears, and turnovers, which we followed when cleaning and aggregating the data.

To quantify roster continuity and transfer activity, we used several **Inside Lacrosse** resources, including the Division I men's transfer portal tracker and the Top 50 players list for the 2025 season. These sources helped us identify key incoming and outgoing players for each program and supported the construction of our returning production and transfer production variables.

Finally, for context on UVA in particular, we consulted LacrosseReference roster analysis articles that describe recent changes in UVA's lineup and returning production. These articles did not directly enter the modeling dataset but guided how we interpreted the forecast for UVA's offense.

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3. Methods

3.1 Target and predictors

Our primary target variable is team goals per game in season ttt, denoted Goals_g. All predictors are constructed using only information from season t−1t - 1t−1 and roster status entering season:

- Lagged performance features: Lag_Goals_g, Lag_OE, Lag_Sh, Lag_ogSh, Lag_Poss, Lag_TO, Lag_ShPos.

- Roster features: Ret_Production, Transfer_Production.

This ensures that our model uses only information available before the season to predict that season's GPG.

3.2 Train, validation, and forecast splits

We split the dataset temporally to mimic real-world forecasting:

- **Training set**: team-seasons with Season in {2021, 2022, 2023, 2024}.

- **Validation set (backtest)**: team-seasons with Season = 2025, used only for out-of-sample evaluation.

- **Forecast set**: team-seasons with Season = 2026, where Goals_g is missing and must be predicted.

This setup prevents information leakage from the future into the past and directly answers the question: “If we had stopped at 2024, how well could we have predicted 2025?” and then “Using all data through 2025, what do we expect for 2026?”

3.3 Model choice: XGBoost regression

We use XGBoost, a gradient-boosted decision tree model, to model the relationship between the 9 predictors and the continuous target Goals_g.

Key reasons for choosing XGBoost:

- It captures nonlinear relationships without manual feature engineering.

- It naturally models interactions (for example, the effect of pace may depend on shot quality).

- It has robust performance on tabular data with relatively small sample sizes.

The R code implements:

```
params <- list(

  objective = "reg:squarederror",

  eval_metric = "mae",

  eta = 0.05,

  max_depth = 3,

  subsample = 0.8,

  colsample_bytree = 0.8

)
```

We trained the model using `xgb.train` with:

- Up to 500 boosting rounds,
- Early stopping after 20 rounds without improvement on the validation MAE,
- A watchlist containing both training and validation sets.

Training logs show MAE decreasing from roughly 1.7 down to approximately 1.22 on the validation set.

3.4 Evaluation metric

We evaluate predictive accuracy using Mean Absolute Error (MAE) on the 2025 validation set:

MAE=1n∑i=1n∣y^i−yi∣,\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} \left| \hat{y}_i - y_i \right|,MAE=n1i=1∑n∣y^i−yi∣,

where yiy_iyi is the actual goals per game for team iii in 2025 and y^i\hat{y}_iy^i is the predicted value.

In the R script, we compute:

```
val_pred <- predict(xgb_model, dval)

val_mae <- mean(abs(val_pred - val$Goals_g))
```

and obtain `val_mae = 1.220185`.

We interpret this as the typical absolute prediction error for a team's 2025 GPG.

3.5 Final forecasting step

After validating the model, we refit it on all known seasons 2022 to 2025:

- Filter `train_full <- df %>% filter(Season <= 2025)`.
- Train `final_model` on all nine predictors using the chosen number of boosting rounds (`best_nrounds`).
- Construct a forecast dataset with `Season == 2026`.

- Predict `Pred_GPG` for each 2026 team:

```
forecast$Pred_GPG <- predict(final_model, dforecast)
```

We then rank all 2026 teams by `Pred_GPG` to obtain offensive rankings.

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4. Results

4.1 Out-of-sample validation performance (2025)

On the 2025 backtest:

- **MAE (validation)**: 1.22 goals per game (rounded).

Team-level GPG typically ranges from roughly 9 to 17. An average error of about 1.2 goals per game means the model is not precise enough to guarantee exact scores for any single team, but it is reliable enough to rank offenses and assign teams to performance tiers (elite, strong, average, below average).

Training logs also show the validation MAE steadily improving before plateauing, which suggests that the model captures meaningful signal without severe overfitting.

4.2 Forecasted offensive rankings for 2026

Using the final model fit from 2022 to 2025, we forecast 2026 goals per game for each team and sort them in descending order to create `Model_Rank`.

Top 12 projected offenses (2026):

Model Rank	Team	Predicted GPG
1	Harvard	14.1

2	Princeton	13.2
3	Syracuse	13.2
4	UNC	12.4
5	Cornell	12.2
6	NotreDame	12.1
7	Georgetown	11.8
8	PennState	11.6
9	John Hopkins	11.2
10	Villanova	11.1
11	Maryland	11.1
12	UVA	11

These teams form the top offensive tier, with projected scoring between roughly 11 and 14.1 goals per game.

We then compared our top 12 offenses in our model to the media's top 12 projected teams from power 4 conferences. There is no media ranking for offense only, so we have to look at the preseason rankings for the entire team in general. We cannot do a direct comparison, but it is helpful for context.

Top 12 projected teams (2026) From Inside Lacrosse

Model Rank	Team
1	Maryland
2	Princeton
3	Syracuse
4	Cornell
5	Georgetown
6	Notre Dame
7	UNC
8	Penn State

9	Duke
10	Ohio State
11	UVA
12	John Hopkins

These teams form Inside Lacrosse. There are no organizations that have top offenses projected

4.3 UVA offensive forecast

For UVA, the 2026 forecast row after prediction contains:

- `Pred_GPG ≈ 11.0`
- `Model_Rank = 12` among 2026 offenses.

In terms of features, UVA's 2026 row has:

- `Lag_Goals_g = 11` (their 2025 GPG).
- `Lag_OE = 31.9`, `Lag_Sh = 25.2`, `Lag_ogSh = 54.3`, `Lag_Poss = 1.3`, `Lag_TO = 54.9`.

These indicate:

- A solid but not dominant recent scoring history.
- Respectable but not elite offensive efficiency and shot volume.
- A relatively high turnover value, which likely drags the forecast down compared to teams with similar raw scoring but better ball security.

Overall, the model places UVA in the upper-middle tier of offenses: competitive with elite programs but slightly behind the very top attacks.

4.4 Feature importance

We compute feature importance using `xgb.importance` on the final model. The top features by Gain are:

Rank	Feature	Gain
1	Lag_Goals_g	0.369
2	Lag_TO	0.195
3	Ret_Production	0.125
4	Lag_OE	0.067
5	Transfer_Production	0.066
6	Lag_ShPos	0.054
7	Lag_ogSh	0.049

8	Lag_Sh	0.038
9	Lag_Poss	0.038

Interpretation:

- Prior goals per game is the single strongest predictor of future scoring, which is unsurprising but reassuring.
- Turnovers are nearly as important. Teams that waste possessions see their scoring suppressed even if they shoot well.
- Returning production and transfers together capture roster continuity and new talent, which strongly influence year-to-year changes.
- Efficiency and shot quality matter above and beyond raw volume and pace.

For UVA specifically, high returning production or strong recent efficiency will tend to lift the forecast, while high turnover levels and lower returning production push it down.

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5. Discussion

5.1 Interpretation of findings

Our results suggest:

- **The model is accurate enough for strategic use.** With MAE about 1.22 GPG on 2025, the model provides a realistic sense of which teams belong in the top offensive tier and which are closer to average.
- **Offensive continuity matters.** Teams that scored a lot last season and bring back their key contributors are rewarded with high projected GPG.
- **Turnovers are a critical lever.** Turnovers emerge as the second-most important variable. Reducing turnovers is an actionable pathway for a team like UVA to outperform its current forecast.
- **UVA is strong but has upside.** A forecast of about 11 GPG and rank 12 indicates a dangerous offense that could jump into the elite tier if a few things break right, such as

improved shot quality, lower turnovers, and strong contributions from transfers.

5.2 Relationship to original analysis plan

Our original analysis plan outlined a broad ecosystem of models: GLMs for wins and goal differential, simulation-based season projections, and potentially expected goals (xG) shot models. In practice, data and time constraints led us to prioritize a focused question, predicting team GPG with XGBoost.

This narrower focus still satisfies key goals of the capstone:

- It uses real-world data to answer a substantive sports analytics question.
- It requires nontrivial model selection, tuning, and validation.
- It produces interpretable outputs with direct relevance for UVA's program and the broader lacrosse community.

5.3 Practical implications for UVA

For coaches and staff, the model suggests:

- **Baseline expectations**: Plan around an offense capable of roughly 11 goals per game against comparable competition.
- **Key focus areas**:
    - Protect the ball by reducing turnovers.
    - Maintain or improve shot quality.
    - Maximize returning and transfer production through roles and schemes.
- **Scenario thinking**: If UVA can shift the underlying features in favorable directions, the model implies a move toward the 12 to 13 GPG range, which historically aligns with deep postseason runs.

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6. Limitations

Key limitations include:

- **Sample size and coverage.** Only 112 team-seasons from a subset of programs are included. Larger coverage across all Division I teams and more seasons would improve stability.

- **Limited covariates.** We do not explicitly model defensive strength, schedule difficulty, injuries, or coaching changes. Some of these effects are indirectly captured via offensive efficiency and prior GPG, but not fully.

- **Point forecasts only.** Our model yields a single expected GPG for each team, but not a full predictive distribution. We can qualitatively discuss best- and worst-case scenarios, but we do not formally quantify uncertainty intervals.

- **Model complexity.** XGBoost is harder to interpret than simpler linear models. We mitigate this via feature importance, but we do not compute SHAP values or partial dependence plots in this deliverable.

- **Dependence on past patterns.** The model assumes that relationships between features and scoring remain stable over time. Structural changes in rules, style of play, or recruiting could reduce predictive power.

- **Unmodeled freshman contributions.** The model does not explicitly account for true freshmen, whose college-level offensive impact is unknown prior to the season. This may bias projections downward for teams expecting immediate contributions from highly touted freshman classes.

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7. Future Work

If we had more time or were continuing the project, we would:

- Expand the dataset to more seasons and all Division I teams, increasing training data.

- Incorporate defensive variables and predict goal differential or win probability directly.

- Use Bayesian or hierarchical models to better share information across similar programs.

- Add shot-level expected goals models so shot quality is derived from individual attempts rather than season aggregates.

- Quantify uncertainty with prediction intervals or bootstrapped simulation, for example by simulating full seasons under varying offensive parameters.

8. Conclusion

We built an XGBoost regression model to forecast team-level goals per game in Division I men's lacrosse, focusing on UVA's 2026 offensive outlook. The model, trained on lagged offensive metrics and roster continuity features, achieves an out-of-sample MAE of about 1.2 goals per game in 2025 and produces a ranked forecast of 2026 offenses.

The results place UVA at approximately 11 goals per game, 12th in offensive strength among teams in our dataset. This is solidly competitive, with clear statistical levers, such as turnovers, returning production, and shot quality, that could move the program into the elite tier.

This work demonstrates how modern machine learning methods, applied carefully and transparently, can inform realistic expectations and strategic decisions in collegiate sports.